HOW EUROPE'S GAS MARKET REFORM CHANGED THE SHAPE OF ITS VULNERABILITY

European debates on energy security are conducted almost entirely in the language of volumes and suppliers: how much gas? from whom? through which pipelines? The policy conclusion is diversification (more interconnectors and more suppliers), but the crisis of 2021 fits this framing badly. Long before Russia curtailed physical deliveries, and months before the invasion of Ukraine in February 2022, European gas prices had already risen several-fold while storage was low but not empty and the pipelines remained open. The damage to households, industry and public budgets arrived through price, not physical shortage. This essay addresses the problem that follows: if European vulnerability now expresses itself primarily through price, then a security strategy built exclusively on supplier diversification is aimed at the wrong target.

Argument: *the liberalisation of European gas markets (a reform designed to reduce supplier power) unintentionally created a new and subtler channel of leverage, operating through the mechanics of price formation rather than through the closing of valves.*

Claim 1: the old contractual order constrained both sides

From the mid-1960s, European gas trade was organised around the Groningen model, named after the Dutch field discovered in 1959. Contracts ran twenty to twenty-five years; prices followed formulas indexed to oil products; and buyers accepted take-or-pay clauses, paying for minimum annual volumes whether consumed or not. This is usually described as a constraint on buyers, and it was. It was equally a constraint on sellers: Högselius (2013) shows that Soviet gas kept flowing westwards through the sharpest crises of the Cold War, including the invasion of Afghanistan and martial law in Poland, because both parties were locked into a structure that was expensive to break. He also concludes that the Soviet "energy weapon" was weak in practice, even though Western Europeans prepared for disruption by building storage and alternative supplies. Coercion was available in principle but costly, visible and legally actionable.

Claim 2: liberalisation relocated price formation from the contract to the hub

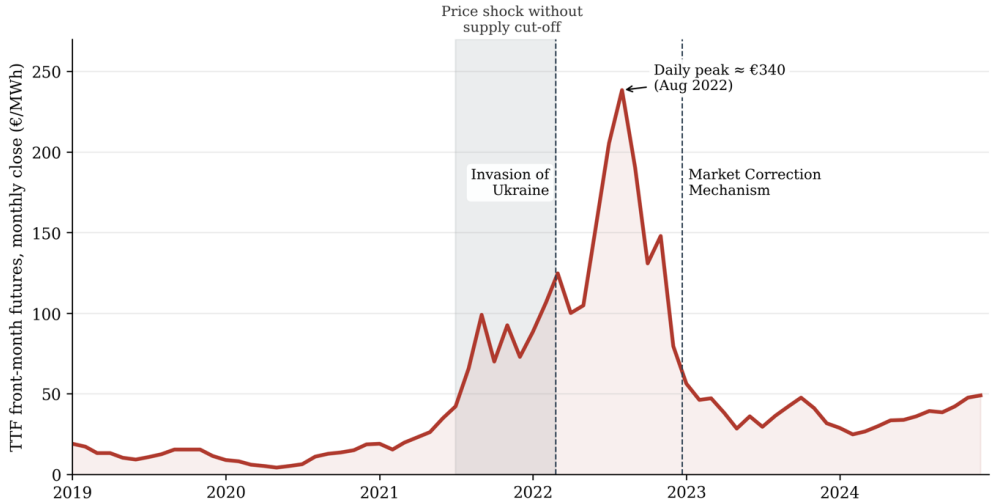
Three European Union gas directives (1998, 2003, 2009) progressively separated transmission from supply and created competitive wholesale markets. Their centre of gravity became the Title Transfer Facility (TTF), a virtual Dutch trading point launched in 2003. Through the 2010s, long-term contracts were renegotiated, often through arbitration, to reference hub prices rather than oil, so that by 2021 most gas in north-western Europe was priced off hubs, with TTF as the benchmark. The critical property of such a market is that it clears at a single price, set by the marginal supply needed to meet demand. When supply is ample, the marginal unit is cheap and competition disciplines suppliers, precisely the intended effect; when supply is tight, the marginal unit becomes extremely expensive, and because heating and industrial demand is inelastic in the short run, the entire market, and every contract indexed to it, reprices upward.

Claim 3: in 2021 this permitted pressure without breach

Through 2021, Gazprom broadly delivered its contracted volumes: what it did not do was offer additional spot volumes or refill the European storage it controlled, notably Rehden in Germany. In September 2021 the International Energy Agency observed that Russia was fulfilling its long-term contracts but exporting less than in 2019, and "could do more" (IEA, 2021). No contract was breached: there was no legal trigger, no casus belli, and no identifiable act of aggression to which governments could respond. Yet because prices were now set at the margin, withholding the marginal molecule was enough to move the entire price level.

By late December 2021, two months before the invasion, TTF front-month prices had exceeded €180 per megawatt-hour, against pre-crisis norms below €20. Coercion had become deniable, cheap and difficult to attribute in real time. The August 2022 peak of around €340 belongs to a different phase, when Russia was openly cutting flows through Nord Stream.

Figure 1. European gas prices, 2019-2024



Source: author's elaboration on Investing.com data (Dutch TTF Natural Gas Futures, front-month, monthly close). The daily peak exceeds the monthly close shown.

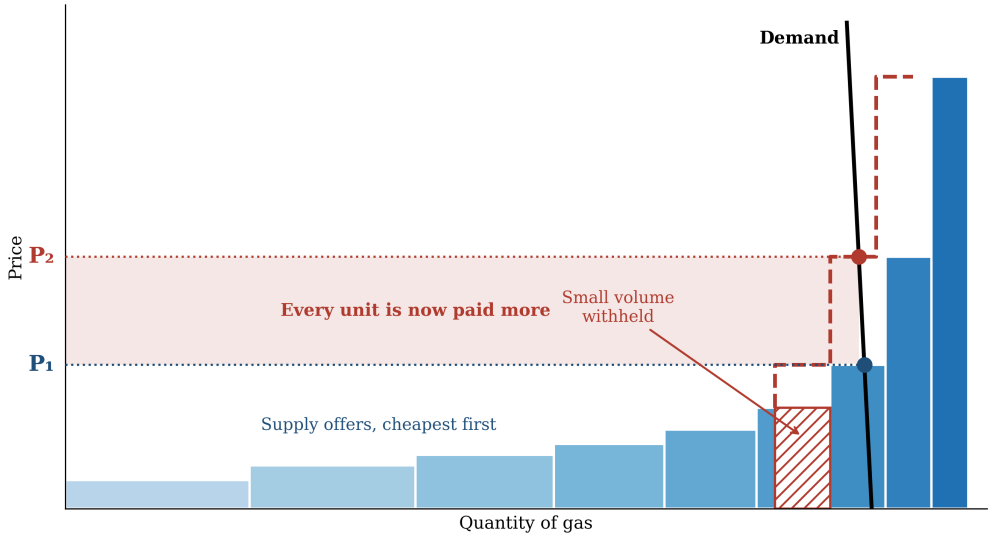
Claim 4: this is weaponised interdependence applied to a price node

Farrell and Newman (2019) argue that when exchange runs through networks, states with jurisdiction over central nodes can turn connection into coercive capacity, through panopticon effects (informational access) or chokepoint effects (the capacity to restrict flows). Their cases, SWIFT, dollar clearing and the internet, are centralised hubs under a single state's jurisdiction. My contribution is to extend this logic to a node that is informational and financial rather than physical or jurisdictional: the TTF is not a pipeline anyone can close, nor an institution any state can command. It converts European scarcity into a single number that governs contracts, electricity prices and fiscal commitments across the continent. Influencing that number requires no control of the node, only influence over the marginal supply it prices. Interdependence was weaponised not at the valve but at the point of price discovery: a chokepoint effect without control of the chokepoint.

Claim 5: the EU's response corroborates the diagnosis

Had the problem been purely one of volumes, the remedies would have been mainly infrastructural. The EU pursued those, through LNG capacity, joint purchasing and demand reduction, but it also intervened in price formation and commercial discretion. Regulation (EU) 2022/1032 imposed binding storage-filling targets and certification of storage operators, removing the discretion that had let strategic storage sit empty. Council Regulation (EU) 2022/1854 capped inframarginal electricity revenues. Council Regulation (EU) 2022/2578 created a Market Correction Mechanism, a conditional cap on TTF front-month prices; it was never triggered, but its adoption shows that the price itself had become something to defend. Each measure treats market design as an instrument of security policy: an implicit admission that market design had become a security vulnerability.

Figure 2. Marginal pricing: a small cut, a large price rise



Stylised illustration. The price is set where demand meets the last offer needed (P_1). Withholding a small volume shifts the remaining offers to the left; with inelastic demand the price jumps to P_2 for the whole market.

Limits and weak points

Three weaknesses deserve acknowledgement. First, the market that transmitted the shock also resolved it: because TTF was liquid and prices high, LNG cargoes were redirected from Asia to Europe within months, a substitution Groningen-era contracts would never have allowed. Liberalisation raised price vulnerability while lowering volume vulnerability. Second, intentionality cannot be demonstrated: the 2021 escalation also reflected demand recovery, weak wind generation and strong Asian LNG demand. The defensible claim is that Gazprom's conduct was consistent with strategic withholding, which market architecture made unusually effective. Third, the concept is stretched: Farrell and Newman's nodes are controlled through jurisdiction, whereas Russia's influence over a Dutch hub was indirect and probabilistic. Moreover, the Russia–Ukraine transit disruptions of 2006 and 2009 show that physical coercion did not disappear with liberalisation. The difference between the two eras lies in method, deniability and attribution cost.

Conclusion

The opening question, "from whom should Europe buy gas?", is incomplete. A continent can diversify suppliers impeccably and remain exposed, because what reaches households is not a molecule but a price, set by whoever can influence the last unit brought to market. Resilience depends on who holds storage, how much supply is contracted outside the spot market, and how the marginal price is formed. This explains an asymmetry in Europe's recovery: physical supply was replaced within a year, yet prices settled at two to three times their pre-crisis level. The infrastructure question was answered faster than the market-design question because it was asked first.

Energy security in a liberalised system is thus partly a question of market architecture: the pipeline determines whether gas arrives, but the design of the market determines what arriving gas can be made to cost.

References

- Council of the European Union (2022a) Council Regulation (EU) 2022/1854 of 6 October 2022 on an emergency intervention to address high energy prices. *Official Journal of the European Union*, L 261.
- Council of the European Union (2022b) Council Regulation (EU) 2022/2578 of 22 December 2022 establishing a market correction mechanism to protect Union citizens and the economy against excessively high prices. *Official Journal of the European Union*, L 335.
- European Parliament and Council of the European Union (2022) Regulation (EU) 2022/1032 of 29 June 2022 with regard to gas storage. *Official Journal of the European Union*, L 173.
- Farrell, H. and Newman, A. (2019) 'Weaponized Interdependence: How Global Economic Networks Shape State Coercion', *International Security*, 44(1), pp. 42–79.
- Högselius, P. (2013) *Red Gas: Russia and the Origins of European Energy Dependence*. New York: Palgrave Macmillan.
- International Energy Agency (2021) *Statement on recent developments in natural gas and electricity markets*, 21 September. Paris: IEA.
- Investing.com (2026) *Dutch TTF Natural Gas Futures: historical data*. Available at: https://www.investing.com (Accessed: 2 October 2026).